FIRST LEGO LEAGUE COACHING PHILOSOPHY

Three Teams, One Philosophy

Persistence, transformation, and accelerated learning

CHAPTER 10 REVIEW EDITION

Coach Ling

For editorial and family review

CHAPTER 10

Three Teams, One Philosophy

The Names in the Auditorium

At an award ceremony, a team name sounds different when it belongs to your children.

At the 2024 FIRST Championship in Houston, Golden Phoenix sat with the other FIRST LEGO League teams in the auditorium. The children had completed their rookie season. Months earlier, some had been too shy to introduce themselves. Now they had presented their robot, project, engineering process, and personal growth on the world stage.

Then we heard the words **Engineering Excellence Award**.

For a moment, we could not believe our ears.

Coach Min Lu led the children toward the stage. I walked with them while recording the moment. They exchanged high-fives, received the trophy, and raised their team flag where the world could see it.

It was epic.

It would also be easy to tell the story badly.

I could describe only the award and make the season sound like a smooth climb from rookie meeting to Houston stage. I could turn shy children into born presenters, an unfinished prototype into a perfect invention, and months of uncertainty into a formula any coach could copy.

That would not help a first-time coach.

Our three teams did not follow one formula:

- Fire Kirin #33274 began in the 2017-2018 *HYDRO DYNAMICS* season and reached the FIRST Championship during *SUPERPOWERED* in 2023.
- Golden Phoenix #61064 began in the 2023-2024 *MASTERPIECE* season, became the Arizona champion, and received the Engineering Excellence Award at the 2024 FIRST Championship.
- Iron Pegasus #72674 began in the 2025-2026 *UNEARTHED* season. It received the Champion's Award and first place in Robot Performance at the ASU Polytechnic regional qualifier, then received the Rising All-Star Award at the 2026 FIRST Championship. Coach Min received the Coach/Mentor Award there.

All three teams advanced to the Arizona state championship in every season they competed. Each reached the FIRST Championship once.



Three world-championship honors, three different stories: Golden Phoenix's 2024 Engineering Excellence Award, Coach Min Lu's 2026 Coach/Mentor Award, and Iron Pegasus's 2026 Rising All-Star Award.

Those facts make the teams look similar. Living through the seasons showed how different they were.

Fire Kirin needed persistence. Golden Phoenix needed transformation. Iron Pegasus needed accelerated learning.

The philosophy remained the same: use robots to build children, let the children own the work, treat the engineering process as more important than a polished answer, and make Core Values visible everywhere.

The coaching could not remain the same.

Fire Kirin: Persistence



Fire Kirin at the 2023 FIRST Championship in Houston. The photograph records one event; the team's real story took six seasons to reach it.

Why Not Now?

Fire Kirin began because a school principal did not accept my cautious timeline.

After judging an FLL competition, I told the principal of Hope Chinese School how exciting the program was and how well it could serve our students. I said we should create a class, but we needed time to prepare.

She asked, “Why not now?”

I agreed with more uncertainty than confidence.

Ten students, ages nine to eleven, registered for the new class. We met for two hours during Hope Chinese School's regular class time in room IRN 125 at Chandler-Gilbert Community College's Pecos campus. On Sundays, when parents could provide a meeting place, we added approximately four more hours. Families shared the registration, robot, field-kit, shirt, and travel costs.

What we did not share clearly enough was the expected commitment.

We had no established team history, no veteran students, and few tools for keeping work moving between meetings. Some children attended inconsistently. Families did not always understand how much work a full FLL season required. Meeting space outside school was difficult to find.

For the *HYDRO DYNAMICS* Innovation Project, the children created an app concept to help households save water and presented it through a fun skit. They built and programmed a robot. They entered their first regional tournament believing they had become a team.

They did not advance.

The Comment I Understood Years Later

During judging, one child said something that suggested the coach had designed the robot.

That was not true. The child had spent less time with the team and did not understand how the work had developed or how a careless sentence could be interpreted. I did not recognize the full significance immediately. Only years later, after gaining judging experience, did I understand that the comment may have created serious doubt about student ownership.

It would be convenient to blame one sentence for the result. The deeper problem was ours.

If one team member could not explain who did the work, we had not involved every child deeply enough. If a child did not know how to speak honestly about the team's process, we had not prepared the entire team for judging. If commitment differed widely, we had not created a clear agreement with families.

After winter break, the few continuing veterans reflected on the season. New children joined later. We began rebuilding not only the robot, but the system around the children.

This story is why I now warn parents and mentors so strongly: **the children must do the work**. The rule protects more than award eligibility. Ownership gives each child a truthful story to tell.

Six Seasons of “Not Yet”

Fire Kirin did not become a championship team the next year.

For several seasons, simply advancing to the Arizona state championship felt important. The team returned without a state-level award. Children graduated or left. New members joined. My daughter was the one student who remained through the full history from the original team to the 2023 FIRST Championship.

We kept improving the structure around each new roster:

- Discord supported communication between meetings.
- Google Classroom organized weekly lessons and announcements.
- Google Drive preserved reports, code, photographs, and team resources.
- Roadmaps and timelines made the season visible.
- An electronic engineering journal recorded what happened after each meeting.
- Spring and the offseason became time for fundamentals; fall became competition season.
- Each season began with the history and lessons of earlier teams instead of a blank page.

We tried tools, kept the ones children actually used, and changed the ones that became adult paperwork.

During the COVID-affected *CARGO CONNECT* season, Fire Kirin competed with only two members and received the Robot Design Award. The following season, the team became the Arizona champion.

The sequence matters.

The first award did not prove that the earlier seasons had been wasted. It revealed what those seasons had accumulated: technical knowledge, routines, documentation, resilience, and a culture that could survive a very small roster.

The Team That Reached Houston

During *SUPERPOWERED*, Fire Kirin investigated vehicle-to-grid technology. The children proposed smart chargers that could allow electric vehicles to support the electrical grid. Their prototype used a Raspberry Pi, an LCD screen, and a 3D-printed enclosure.

The robot completed reliable missions. The children could explain why it worked.

My daughter, then studying AP Calculus in eighth grade, could describe PID control in her own words and connect it to gyro-straight movement, gyro turns, and line following. A nine-year-old rookie served as the chief robot builder, a fact that surprised judges who might have expected the oldest child to own the most technical role. All four team members were strong presenters.

The team advanced to the 2023 FIRST Championship in Houston.

There, Fire Kirin met international teams with nearly perfect Robot Game runs. Some matches required a fourth attempt to break ties. The children saw impressive prototypes, exchanged ideas, made friends, and discovered that the world stage was not a finish line. It was a much larger classroom.

One member returned convinced that our robot needed acceleration and deceleration control to move faster without sacrificing accuracy. The idea was implemented during *MASTERPIECE* and refined through *UNEARTHED*. A lesson observed at one championship became knowledge inherited by children who had not attended it.

That is how a mature team should use Houston.

Success Can Empty a Team

After Houston, almost every original Fire Kirin member left. Some had aged out. Some pursued other programs. Some felt they had already reached the highest stage available to them.

Only my daughter was willing to remain long enough to teach newer members before moving into high-school robotics.

We had achieved the result teams dream about and lost much of the team that achieved it.

This is one of the uncomfortable truths of youth programs: a major success can remove the goal that held a roster together. A team that never reaches Houston may remain hungry and keep learning together. A team that reaches it may decide the story is complete.

Fire Kirin had to restart.

Its history still mattered, but history could not operate a robot or welcome a rookie. The coaches had to help new children inherit the name without demanding that they reproduce the old roster.

Persistence does not mean keeping the same children forever. It means keeping the learning available when the children change.

Golden Phoenix: Transformation



Golden Phoenix at a 2025 SUBMERGED regional qualifier. The roster continued evolving after the team's rookie trip to Houston.

Children Who Could Barely Say Their Names

Golden Phoenix began with an advantage Fire Kirin never had: it could learn beside a veteran team.

It also began with an entire roster of rookies.

At the first meeting, most members were shy. Shiyu and his twin brother, Peiyu, could barely introduce themselves. Other children were more outspoken, including one younger brother whose excitement created a simple goal for the team: “Go to Texas.” The phrase became part of their judging-room presentation.

Rookies did not carry the pressure of defending Fire Kirin's Arizona championship. They had no old score to protect and no personal history of how things “should” be done. Their weakness was inexperience. Their freedom was the same thing.

During the *MASTERPIECE* season, Golden Phoenix did not lead the Arizona Robot Game standings. It still became the Arizona champion.

The children showed judges a more complete picture of what they had learned.

An Honest Prototype

Golden Phoenix's Innovation Project explored how deaf people might experience music. The children produced design drawings and interviewed a school principal, teachers, and deaf students. They learned from people connected directly to the problem.

They did not arrive at Houston with the kind of polished prototype that fills a crowded pit with light, motion, and confidence.

Before the championship, the mentors demonstrated how SPIKE Prime parts could be used for prototyping. We showed examples from past projects to spark ideas. The children produced preliminary work, but the prototype remained incomplete.

The adult temptation was obvious: fix it.

An experienced adult could have constructed something shinier, rehearsed an explanation around it, and made the table look more competitive. It also would have weakened the most important fact in the room: the children owned what they brought.

They showed the judges their real work and explained what they hoped to do next.

An unfinished student prototype is stronger evidence than a finished adult one.

“I Don't Think You Are Shy”

At the 2024 FIRST Championship judging session, a judge asked each child what FLL had taught them.

Shiyu explained that he had been very shy at the beginning of the season but had learned to speak in front of strangers.

The judge replied, “I don't think you are shy.”

That response proved his point.

Within one rookie season, a child who had struggled to say his name at the first meeting was telling a judge at the world championship how the program had changed his personality. In later seasons, he became a team leader.

His growth was not separate from the technical work. When given ownership, he could teach acceleration and deceleration programming together with PID-controlled movement-one of the team's new technical challenges. He also learned to recognize what a judging conversation still needed. When judges asked whether the team had anything else to share, he helped the group remember to explain its Core Values.

Technical confidence and speaking confidence grew together because the team gave him something real to explain.

Engineering Excellence Without Pretending to Be Perfect

The 2024 championship used a separate pit interview for Robot Design. Children could demonstrate the robot and present beside their pit in a format that felt more like an FRC or FTC interview.

Golden Phoenix showed CAD designs on a tablet. The children explained prepared control algorithms and testing charts. Their engineering documents made the process visible.

In the judging room, Innovation Project and Core Values reinforced each other. The project was authentic. The stories of personal growth were authentic. The children did not need to perform the role of experienced champions. They needed to show what rookies had genuinely learned and how they had worked together.

Then their name was announced for the Engineering Excellence Award.

Some children continued with Golden Phoenix in later seasons. Some left to pursue other robotics opportunities. The team's rookie story became part of our program history and changed what the next rookies believed was possible.

Golden Phoenix transformed more than its own members.

When Iron Pegasus formed, “rookie team” no longer sounded like a reason to lower expectations. The children had evidence beside them that a first-year team could reach the world stage.

Iron Pegasus: Accelerated Learning



Iron Pegasus with Coach Min Lu at the 2026 FIRST Championship in Houston.

Three Weeks Before Regionals

Iron Pegasus was not created by coaches who had forgotten how hard a rookie season could be.

It was created inside a program with Fire Kirin's history, Golden Phoenix's example, veteran teaching, past robots, reports, code, experts, and competition routines. That inheritance allowed the children to learn faster. It did not allow them to skip the work.

Three weeks before the ASU Polytechnic regional qualifier, the team's reports were incomplete, the skit was unstable, the operators were inconsistent, and the Robot Game was not ready.

During one weekend meeting, Coach Min and I asked the children to perform the Innovation Project skit.

They could not remember their lines. The words they did remember sounded recited. Their explanation of four board-game iterations was so disorganized that an audience could not see how one design had led to the next.

We did not rewrite the project for them.

I asked four children to hold the four versions of the game map. As each version was discussed, the child unfolded it while the others explained what had changed and why. The physical sequence gave structure to the engineering story.

At the next rehearsal, the explanation was visibly clearer.

This is what accelerated learning looks like in practice. It is not an adult supplying a faster answer. It is an adult finding a demonstration that allows children to reorganize their own work.

Different Children Need Different Doors

Iron Pegasus included a neurodivergent veteran whose attention, communication, and technical focus did not always look conventional.

The child might become distracted during judging, miss a deadline, or switch tasks before completing the first one. The same child could become deeply absorbed in programming and LEGO mechanisms, stay late to refine an attachment, and notice possibilities others missed.

The coaching response was not to define the child by either the difficulty or the talent.

We gave one clear instruction at a time. We reduced distractions. We focused on one mission before moving to another. Before competition, we communicated relevant needs to tournament leadership so the judging environment could be supportive.

As the team's veteran, the child already understood coding, attachments, and lessons from the previous season. The leadership challenge was to stop solving everything personally. The veteran began asking younger teammates to own missions, then guiding them instead of writing all the code.

Inclusion is not lowering the engineering standard. It is designing a path through which a particular child can reach it-and contribute something the team would otherwise lack.

Four Versions of Archaeology

For its *UNEARTHED* Innovation Project, Iron Pegasus asked how people could learn what archaeology really is, beyond the popular image of treasure hunting and *Indiana Jones*.

The children researched the market and found few educational board games focused on authentic archaeology. They consulted archaeologists, developed a question bank, and iterated the game through four versions. They

explored a website and digital versions, using AI tools as part of broadening access. They also contacted an archaeology organization about directing support toward archaeological study.



Iron Pegasus made iteration visible at its Houston pit: successive physical game boards led to a later digital version.

The project was not impressive because an adult could describe all those components in one paragraph.

It became impressive when the children could place four maps in front of an audience and say:

This is what we tried. This is what we learned. This is what we changed.

That sentence is the heart of the engineering design process.

From Unstable to Rising All-Star

At the regional qualifier, Iron Pegasus's final Robot Game attempt raised the team into first place in Robot Performance. The team also received the Champion's Award.

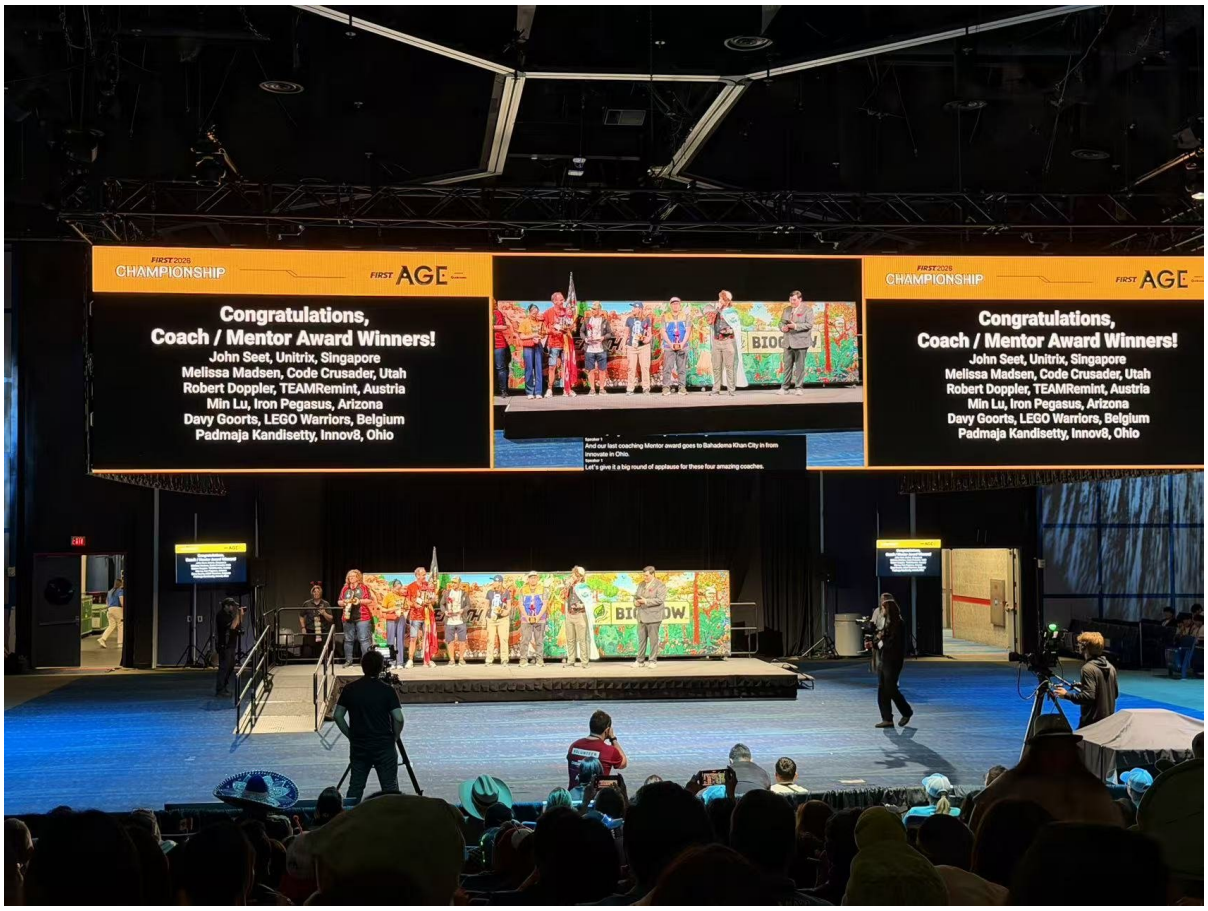
Coach Min had watched the judging session and knew the children had done strong work. We were happy, but not shocked. They had earned the result through the repetitions that the ceremony did not show.

The state-championship rubric still identified uneven speaking and participation. Some children carried more of the explanation than others.

By the 2026 FIRST Championship, every team member spoke. Children complemented one another's answers instead of waiting for one star presenter. The team received the Rising All-Star Award.

Our interpretation was not that the children had suddenly become polished. They had become more visibly a team.

The Coach/Mentor Award for Min was a genuine surprise. Team members and parents had nominated her. The judges saw perspectives that coaches do not often hear directly: how she organized the team, responded to families, cared for each child, followed the details, and planned competition day like clockwork.



Coach Min Lu onstage with the 2026 FIRST Championship Coach/Mentor Award recipients. The nomination came from team members and parents.

The award mattered because it came through the appreciation of the people her work supported.

It also returned the attention to the adults' real responsibility. A coach does not create a Rising All-Star by standing in the spotlight. A coach creates enough order, safety, challenge, and belief for children to step into it.

Same Philosophy, Different Coaching

These stories are not three proofs that our method always produces trophies.

Fire Kirin attended Houston without receiving an award. Its members still became pioneers who opened a larger world for every team that followed. Golden Phoenix received a major award with an unfinished prototype. Iron Pegasus reached regionals with its work still unstable and needed much more direct demonstration than our older teams.

The philosophy stayed consistent. The assignments, feedback, and amount of structure changed.

Team	What the children most needed	What the coaches had to practice	What success taught the program
Fire Kirin	Time, continuity, systems, and permission to keep going	Patience across seasons; turning mistakes into reusable knowledge	“Not yet” can last for years without meaning “never”
Golden Phoenix	Opportunities to speak, authentic ownership, and freedom from veteran expectations	Stepping back from an imperfect prototype; seeing transformation as evidence	Rookies can do ambitious work without pretending to be veterans
Iron Pegasus	Visible examples, small steps, frequent feedback, and differentiated support	Demonstrating without taking over; matching instruction to each learner	Inherited knowledge can accelerate learning when veterans build independence

Fair coaching did not mean giving every team the same assignment.

An assignment that stretches a veteran may overwhelm a nine-year-old rookie. A long verbal explanation may work for one child and disappear for another. One team may need a principle and room to explore. Another may need the coach to unfold the first map and ask what should happen next.

The boundary is not “hands off.”

The boundary is ownership.

We can show the rope, demonstrate key concepts, spark interest, ask questions, organize feedback, and create a safer learning environment. The children must make the decisions, conduct the work, understand the result, and tell the story.

What the Teams Gave One Another

The three teams met in the same classroom. Their results could have created status: the old champion, the world-award rookie, and the youngest team.

We tried to create inheritance instead.

Veterans taught rookies. Teams watched one another's judging rehearsals and Robot Game runs. Children celebrated another roster's award because they understood that the achievement expanded what their whole program believed possible.

Technical ideas crossed team boundaries with credit. An Iron Pegasus veteran designed an attachment in which one motor drove two arms at different speeds through different gear ratios. The idea inspired a Golden Phoenix builder, who developed a different final design. The value was not in copying a mechanism. It was in seeing a peer ask an engineering question the other child had not considered.

The exchange also moved beyond Arizona.

At the world championship, Golden Phoenix competed beside the Spanish team Legotronic Beavers. The Spanish children cheered for Golden Phoenix at the Robot Game table and lifted one of our members afterward. The moment became one of our clearest examples of Gracious Professionalism: teams can compete seriously and still increase one another's joy.

Houston should not be treated as a private reward for one roster.

Before going, ask experienced coaches about travel, packing, pit supplies, schedules, and outreach. During the event, visit other pits. Let children ask children about robots, projects, team culture, and mistakes. Exchange ideas and make friends. After returning, reflect and teach what the team learned to people who were not there.

An experience becomes program knowledge only when someone passes it on.



Three team identities beside a FIRST Alumni Volunteer patch: the cycle is complete when students return to create opportunities for others.

A Coach's Case-Study Questions

When you study a successful team-ours or anyone else's-do not begin with “How did they win?”

Ask:

1. What could the children not yet do at the beginning?
2. What did the adults teach, and what did they deliberately leave unfinished?
3. Which team systems made progress possible between meetings?
4. How did quieter, younger, or differently wired children gain meaningful ownership?
5. What evidence shows iteration rather than a polished final result?
6. What failure changed the next season?
7. What knowledge remained after the original roster left?
8. Who did the team help beyond itself?

Awards can be useful evidence, but they are incomplete evidence.

A trophy cannot show the first meeting when a child could not say his name. It cannot show the years when a team advanced without an award, the Sunday a skit collapsed, the prototype adults chose not to finish, or the veteran who learned to hand a mission to a rookie.

Those are the moments a coach can reuse.

Three Messages to a First-Time Coach

Fire Kirin says: **Do not wait until you feel fully prepared. Start, reflect, and persist.**

Golden Phoenix says: **Do not confuse inexperience with low potential. Rookies can reach a world stage while doing authentic rookie work.**

Iron Pegasus says: **Do not make every new child rediscover everything alone. Let veterans teach, then help them step back so rookies can own the next solution.**

Together, the teams say something more important:

Core Values must live in the program's DNA. We use robots to build children. We broaden the meaning of winning so children can beat yesterday, learn from a loss, and return knowledge to the people who come next.

I hope our alumni remember the scores, flags, and auditoriums. I hope they remember even more clearly the first time they found their voice, taught another child, recovered from failure, or discovered that their mind could solve a difficult problem.

Someday, I would like to see a documentary that places those childhood moments beside the adults the children become.

Until then, these three team histories offer the evidence I can give a new coach:

You do not need to produce a miracle.

Create the conditions in which children can do work that surprises you.

Then let the story belong to them.